IMAGES

Mirror, Water and Fold-Line Images

OAVS TGT | Reasoning (Non-Verbal) | Topic 14: fifth non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, Rule Detection, **Images**, Completion of Pattern, Mirror Images, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). **No question labelled ‘Images’ appears in these papers.** The two nearest PYQs, English 2019 Q47 and Q49 (‘how would the pattern appear when the transparent sheet is folded at the dotted line’), are image-by-folding questions and are solved here. Mirror-image questions (8 PYQs) get their own PDF next.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	What is and is not in the papers; index of the 2 PYQs
2	Foundation	Mirror image, water image and fold-line image; symmetric letters and digits
3	Solved PYQs	English 2019 Q47 and Q49 with their original figures
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	11 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Images by folding: 2 PYQs, both in English 2019

Paper	Q. numbers	Count	What was asked
English 2019	47, 49	2	How would the pattern appear when the transparent sheet is folded at the dotted line
English 2022	none	0	—
Mathematics 2024	none	0	—
Science 2024	none	0	—
Social Studies 2023	none	0	—
Total		**2**	**Both in English 2019**

PYQ INSIGHT: **Honest picture:** your five papers contain no water-image question and no question labelled ‘images’. Both transparent-sheet questions are about the **reflection across the fold line**, so reflection is the idea to master. The mirror-image questions of the papers (8) come in the next PDF.

Pattern	PYQs	Priority
Fold along a line: the folded half is reflected and overlaps the other half	2	★★★ Must
Expected: water image of letters, numbers and words	0	★★ Expected
Expected: mirror image of letters and words (covered in detail in the next PDF)	0	★★ Expected

Index of the 2 PYQs

Paper / Q	Question in short	Ans	See
Eng 2019 Q47	Diagonal dotted line; circle in one half, arrow in the other; fold at the dotted line	Option 2	3.1
Eng 2019 Q49	Vertical dotted line; circle and triangle on the left, star and inverted triangle on the right; fold at the dotted line	Option 2	3.2

CHAPTER 2

Foundation

Three kinds of images

Image	Mirror (axis)	What changes	Example (word RAT)
Mirror image (mirror to the right or left of the figure)	Vertical line	Left and right swap: the **order of letters is reversed** and each letter is flipped left-right	TAR written back to front
Water image (mirror below or above)	Horizontal line	Top and bottom swap: each letter is flipped upside down; **order stays the same**	RAT upside down, same order
Fold-line image (transparent sheet folded)	The dotted line	Each shape on one half is reflected across the line and lands on the other half	Q47, Q49

2.1 Reflection rules

- A reflected point is the **same distance** from the line, on the other side.
- An arrow that points **towards** the line points **away** from it in the image; the component along the line does not change.
- A **vertical** line swaps left and right ($x \to -x$). A **horizontal** line swaps up and down ($y \to -y$).
- A **diagonal** line (bottom-left to top-right) swaps x and y: up-left becomes down-right.
- A shape symmetric about the line looks unchanged; an asymmetric shape (letter F, arrow, R) shows the reversal clearly.

2.2 Symmetric letters

Axis	Letters that look the same	Digits
Vertical axis (mirror image)	A, H, I, M, O, T, U, V, W, X, Y	0, 8 (1 in many fonts)
Horizontal axis (water image)	B, C, D, E, H, I, K, O, X	0, 3, 8
Both	H, I, O, X	0, 8

SHORTCUT: Quick test for words: a word looks the same in its water image if **every** letter is symmetric about the horizontal axis (BOX, DICE). A word looks the same in its mirror image only if it reads the same backwards **and** every letter is symmetric about the vertical axis (TOT, MOM, WOW).

2.3 Fold-line images: method

Step	Do this
1	Decide which half moves and which half stays (usually the half **without** the final answer boxes).
2	Reflect every shape of the moving half across the dotted line: position **and** orientation.
3	Overlap the reflected shapes with the shapes already on the other half.
4	Compare with the options: shapes in the right places, turned the right way.

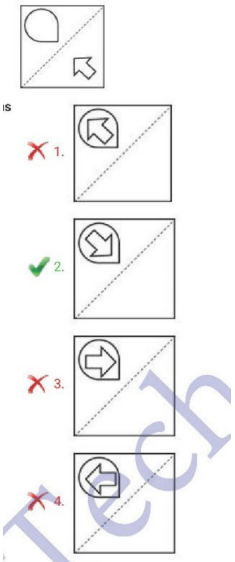
CHAPTER 3

Solved PYQs

Both questions with their original figures

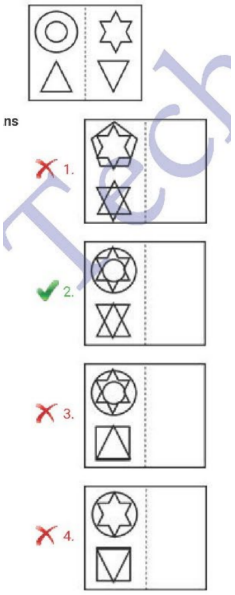
3.1 English 2019 Q47: diagonal fold

How would the pattern appear when the transparent sheet is folded at the dotted line?



1. The sheet is a square with a dotted diagonal from bottom-left to top-right. The upper-left half has a circle (with a pointed corner). The lower-right half has a hollow arrow pointing **up and to the left**.
2. Folding along the diagonal, the lower-right half lands on the upper-left half. A diagonal line (bottom-left to top-right) swaps x and y, so a point at the lower right goes to the upper left (into the circle).
3. The arrow's direction changes: up-left $(-1, +1)$ becomes $(+1, -1)$, that is, **down and to the right**.
4. So the result is the circle with an arrow pointing down-right inside it. Option 1 (arrow up-left) is the unreflected arrow; options 3 and 4 point right and left, which a diagonal fold cannot produce.
Answer: Option 2

3.2 English 2019 Q49: vertical fold



1. The sheet has a vertical dotted line. Left half: a circle (a ring with a smaller circle) at the top and an upright triangle at the bottom. Right half: a six-pointed star at the top and an inverted triangle at the bottom.
2. Folding the right half over the left half reflects the right half left-to-right: the star lands on the circle; the inverted triangle lands on the upright triangle.
3. The star is symmetric, so it stays a star: **circle with a star over it**. An inverted triangle over an upright triangle makes a **six-pointed star** (two triangles).
4. Option 2 shows the circle with the star at the top and the two overlapping triangles at the bottom. Option 4 has a plain inverted triangle and no overlap; option 3 has a square; option 1 has a pentagon-like outline.
Answer: Option 2

SHORTCUT: When the shapes on the moving half are symmetric about the fold line, only their positions matter. Look at **which asymmetric shape** (arrow, flag, letter) decides the answer, as in Q47.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Vertical axis:** left and right swap. **Horizontal axis:** up and down swap. **Diagonal:** swap x and y.
- **Mirror image of text:** reverse the order of letters and flip each letter left-right.
- **Water image of text:** keep the order; flip each letter upside down.
- **Fold-line questions:** look only at the asymmetric parts; symmetric parts stay.
- **Eliminate** options with the wrong order of letters first, then by flipped letters.

Trap	How to avoid it
Mixing mirror and water images	Mirror: vertical axis, order reversed. Water: horizontal axis, order kept.
Reflecting positions but not orientations	An arrow pointing up-left becomes down-right under a diagonal fold.
Forgetting overlap in a fold	Both halves end up on top of each other: draw both.
Using symmetric letters only to decide	Check the asymmetric letters; they show the flip.
Treating a rotation as a reflection	A 180° turn is not the same as a left-right flip (for asymmetric shapes).

Scope

Related PYQs	Where
Mirror images: Eng 2019 Q34, Q46; Eng 2022 Q35; Maths 2024 Q77, Q79, Q93; Science 2024 Q97; SST 2023 Q4	Mirror Images (next PDF)
Eng 2019 Q48; Eng 2022 Q31 (fold, cut or punch, then open)	Paper Folding (coming)

CHAPTER 5

Practice Set

11 original questions; answers on the next page (every option set verified)

Part A: Mirror and water images of text (mirror placed to the right for mirror images)

Q1. Select the correct mirror image (mirror on the right) of **J4kR**.

(A)	(B)	(C)	(D)
ꓶ4ʞꓤ	ꓤʞ4ꓶ	ꓶ4ʞꓤ	Rk4J

Q2. Select the correct water image of **R2GP**.

(A)	(B)	(C)	(D)
ꟼƓƧꓤ	ꟼƓƧꓤ	R2GP	ꓤƧƓꟼ

Q3. Select the correct mirror image (mirror on the right) of **7aSL**.

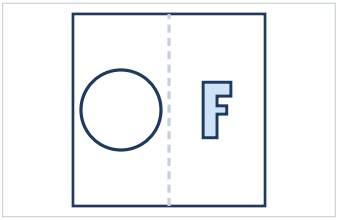
(A)	(B)	(C)	(D)
⅃ᵍƧ⅂	⅂Ƨɐ7	7ɐƧ⅂	LSa7

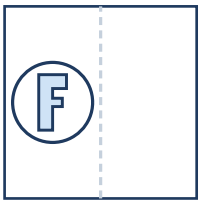
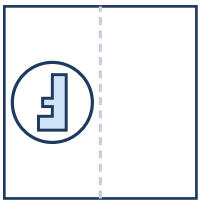
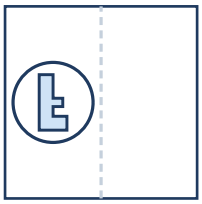
Q4. Select the correct water image of **d3FQ**.

(A)	(B)	(C)	(D)
ϘꟻƐp	ϘƐꟻб	d3FQ	ϘꟻƐb

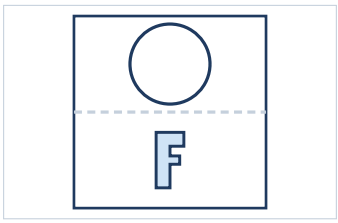
Part B: Folding a transparent sheet at the dotted line

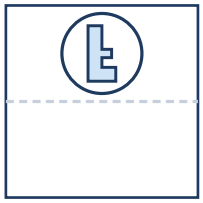
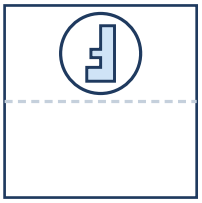
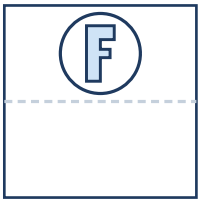
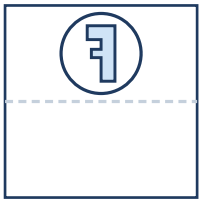
Q5. How would the pattern appear when the transparent sheet is folded at the dotted line?



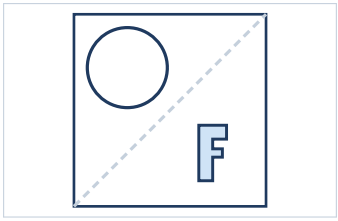
(A)	(B)	(C)	(D)
			

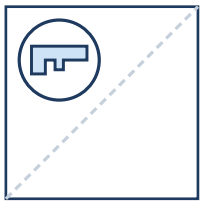
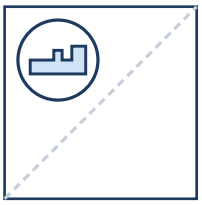
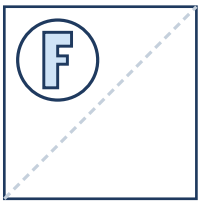
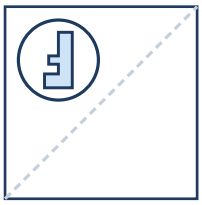
Q6. How would the pattern appear when the transparent sheet is folded at the dotted line?



			
(A)	**(B)**	**(C)**	**(D)**

Q7. How would the pattern appear when the transparent sheet is folded at the dotted line?



			
(A)	**(B)**	**(C)**	**(D)**

Part C: Words and symmetry

Q8. Which word looks the same in its **water image** as the word itself? (A) BOX (B) TOP (C) CAT (D) WAS

Q9. Which word looks the same in its **mirror image** (mirror on the right) as the word itself? (A) TOP (B) TOT (C) BOT (D) POT

Answer key

Q	Answer	Reason
1	(B)	The mirror image reverses the order and flips each letter left-right; (reversed letters unflipped) and (flipped letters in the same order) are the common wrong options.
2	(D)	The water image keeps the order and flips each letter upside down.
3	(B)	The mirror image reverses the order and flips each letter left-right; (reversed letters unflipped) and (flipped letters in the same order) are the common wrong options.
4	(B)	The water image keeps the order and flips each letter upside down.
5	(A)	Vertical fold: the right half is reflected left-right (the glyph is mirrored) and lands on the circle. Wrong options: the glyph unreflected, turned 180°, or reflected across the other axis.
6	(A)	Horizontal fold: the lower half is reflected up-down (the glyph is flipped upside down) and lands on the circle. Wrong options: the glyph unreflected, turned 180°, or reflected across the other axis.
7	(B)	Diagonal fold: x and y swap; the glyph is reflected across the diagonal and lands on the circle. Wrong options: the glyph unreflected, turned 180°, or reflected across the other axis.
8	(A) BOX	B, O and X are all symmetric about the horizontal axis.
9	(B) TOT	TOT reads the same backwards, and T and O are symmetric about the vertical axis.

CHAPTER 6

Quick Revision

One page for the last day

Question	Rule	Check
Mirror image (text)	Reverse the order; flip each letter left-right	Same letters if symmetric: TOT, MOM, WOW
Water image (text)	Same order; flip each letter upside down	BOX, DICE look the same
Vertical fold line	Reflect left-right	Eng 2019 Q49 (star lands on circle)
Horizontal fold line	Reflect up-down	—
Diagonal fold line	Swap x and y; up-left becomes down-right	Eng 2019 Q47 (arrow down-right)

ROUTINE: Exam-day routine: name the axis; reflect position and direction; overlay the two halves; eliminate options by the asymmetric shape. 40 seconds per question. Next in the syllabus order: Completion of Pattern, Mirror Images, Figure Matrix, Paper Folding.